Pledgement

"On my honor, I have neither given nor received any unouthorized and/or inoppropriate ossistance for all sessions of this exon. The work done on this exom is totally my own I understand that by the school code, violation of these principles will leod to a zero grade and is subject to harsh disciplne issues."

150719578
Umuthon KAPTAN

Umuthan KAPTAN
150719578

Q1

a)

$$A=\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$A=\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 2 & 28 & -12 & 4 \end{bmatrix}$$

$$A=\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -15 & 4-\alpha^2 \end{bmatrix}$$

$$A=\begin{bmatrix} 2 & 2\alpha & -2y & 2\alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -12 & 4 \end{bmatrix}$$

$$A=\begin{bmatrix} 0 & 2\alpha-\beta & -2y-3 & 2\alpha-\alpha^2 \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -12 & 4 \end{bmatrix}$$

$$A=\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 28-\beta & -15 & 4-\alpha^2 \end{bmatrix}$$

so

$$A=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$

$K = 28 - \beta$

$\beta = -1$

$\alpha = 0, \quad -y = 0, \quad \alpha = 0 \quad m = 0$

$k = 28 + 1 = 29$

Umuthan KAPTAN
150719578

Q1

(b) Upper triangle Matrix B

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & 6 & 2 \end{bmatrix} \rightarrow U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 10 & 0 & 2 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix} \leftarrow U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0-1 & 10-14 & 0-6 & 0-2 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & -4 & -6 & 0 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & -18 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

Upper triangular matrix

(c) $$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & -1 & 3 & 0 \\ 0 & 29 & -15 & 4 \end{bmatrix}$$

Umuthan KAPTAN
150719578

② $x = 5y$

(a) This can be written

$$x - 5y = 0$$

if it is a four dimensial geometry

there must be 2 other voricble

a and b

$$k = \left\{ \begin{bmatrix} 5 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(b) ronk $k(A) = 4$,

$Ax = B$ unknoun geometry to hove solution

$$k = \begin{bmatrix} 5 & 5 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(c) $x + 4y - z = 6$

$m = 1$, $n = 9$, $0 = -1$

$Ax = b$, hove no solution

$x \leftarrow R^5$ for $Ax = B$

The left null space

A hos dimension inside

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 0 \\ 6 \end{bmatrix}$$

There are 5 dimensial space

Umuthan KAPTAN
150719578

Q3

(a) $H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$

$= I^2 - I \cdot 2uu^T - 2uu^T \cdot I + 4uu^T \cdot uu^T$

$= I(I - 2uu^T - 2uu^T) + 4u \cdot \|u\|^2 \cdot u^T$

$= I - 2uu^T - 2uu^T + 4u \cdot \|u\|^2 \cdot u^T$

As $\|u\|^2 = u^T \cdot u$

i.e $\|u\| = \pm\sqrt{u^T u}$

$= \|u\| = 1$

so, $H^2 = I - 4uu^T + 4uu^T$

$\therefore \boxed{H^2 = I}$

(b) $HH^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T) \cdot (I^T - (2uu^T)^T)$

$= (I - 2uu^T) \cdot (I - 2(u^T)^T u^T)$

$= (I - 2uu^T) \cdot (I - 2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T \cdot uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$

$= I - 4uu^T + 4uu^T$

$HH^T = I$

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

similarly, $H^T H = I$

so, $HH^T = H^T H = I$

so, it is an orthogonal matrix

Q3

Umuthan KAPTAN
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(c) For a matrix A, a vector (non zero) will be eigen vector (suppose, v) of A corresponding eigen value λ if, $\boxed{Av = \lambda v}$

now, $Hv = (I - 2uu^T) \cdot v$

$= Iv - 2uu^T v$

$= v - 2u\|u\|^2$ [As I is identity matrix]

As I is a identity matrix, so I is the only eigen value of I, and $Iv = I \cdot v = v$ (also), and $\|u\| = +\sqrt{u^T u}$, where, u is a unit vector, so, $\|u\| = 1 = +\sqrt{u^T u}$ and hence, $u^T u = \|u\|^2 = 1$

hence,

$Hv = v - 2u \cdot 1 = -v \rightarrow Hv = -v = (-1)v$.

So we have $\Rightarrow Hv = \lambda v$, whose $\lambda = -1$

so, v is an eigen vector of H and corresponding eigen value for v is $= \lambda = -1$.

(d) Given, $H = (I - 2uu^T)$

As, $H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$ $\quad (AB)^T = A^T \cdot B^T$, $((AB)^T = B^T \cdot A^T)$

$= I - 2(u^T)^T u^T$

$= I - 2uu^T = H$

So, H is a symmetric matrix, and there is a theorem which states that,

$H = Q \Omega Q^T$ (where, Q is an orthogonal matrix)

$= Q \Omega Q^{-1}$ (As, $QQ^T = I \Rightarrow Q^T = Q^{-1}$)

$= (Q^{-1})^{-1} \Omega Q^{-1}$ (Ω = Diagonal matrix consist of eigen values of H)

$= P^{-1} \Omega P$

So, H is diagonalizable matrix

Umuthon KAPTAN
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(a4) $v(0) = 10 \quad w(0) = 40$

(a)

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

$v_{k+1} = -2v_k + w_k$
$w_{k+1} = v_k - 2w_k$

differential equation set express the process

(b) after 1 sec, in tank 1

$v_{1+1} = -2v_k + w_k$
$v_2 = -2 \times 10 + 1$
$= -20 + 1$
$= 19\ L \longrightarrow \% = \frac{19}{10} \times 100 = \underline{190\ \%}$

(c) at steady state
25 liters remaining